

7 Induction and Recursion

Definition 7.1: The Unique Existential Quantifier.

For every wff φ with at most one free variable, we define what it means to say *there exists a unique*¹ object that φ is *true* about below.

$$\exists! x(\varphi(x)) \quad :\Leftrightarrow \quad \exists x(\varphi(x) \wedge \forall y(\varphi(y) \Rightarrow y = x))$$

Given a set X , we can extend this definition in a similar way to *restricted quantifier notation*² to make complicated sentences more convenient to express.

$$(\exists! x_1, \dots, x_n \in X)(\varphi(x)) \quad :\Leftrightarrow \quad \exists! x_1 \dots x_n ((x_1 \in X \wedge \dots \wedge x_n \in X) \wedge \varphi(x))$$

¹Notice that the word “unique” means “only one”; this is a *unary* predicate that says x is the *only* thing with some given property. It *does not* mean the same as the word “distinct,” which is a *binary* predicate indicating that $x \neq y$.

²Definition 6.1 from *clavicula* 6

7.1 What is a Function?

Definition 7.2: Function.

Let X and Y be sets. We define what it means for f to be a *function from X to Y* below.

function

$f : X \rightarrow Y$

$$f : X \rightarrow Y \quad :\Leftrightarrow \quad f \subseteq X \times Y \wedge (\forall x \in X)(\exists! y \in Y)((x, y) \in f)$$

We use the notation $f : X \rightarrow Y$ to indicate that f is a function from X to Y .³ The set X in this definition, called the *domain* of f , is the set of all inputs to f ; we will use the notation $\text{dom}(f)$ to denote the domain of f . The set Y contains all of the *possible outputs* of f ; we call this the *codomain* of f and use the notation $\text{cod}(f)$ to denote it. When f is a function from X to Y , we define the following convenient notation for every $x \in X$ and $y \in Y$.

$\text{dom}(\cdot)$

$\text{cod}(\cdot)$

$f(x) = y$

$$f(x) = y \quad :\Leftrightarrow \quad (x, y) \in f$$

Here is a basic, but important, example of a function. Consider $\text{id}_{\mathbb{N}} : \mathbb{N} \rightarrow \mathbb{N}$ given by $\text{id}_{\mathbb{N}}(x) := x$ for every $x \in \mathbb{N}$. Below are some of the first few values of this function.

$$\text{id}_{\mathbb{N}}(0) = 0$$

$$\text{id}_{\mathbb{N}}(1) = 1$$

$$\text{id}_{\mathbb{N}}(2) = 2$$

$\vdots$

identity
function

In fact, for any set A , we can define a version of this function—called the *identity function on A* —denoted by id_A . We define $\text{id}_A : A \rightarrow A$ such that $\text{id}_A(a) := a$ for every $a \in A$.

In order for f to be a function from X to Y , every element $x \in X$ needs to have exactly one corresponding element $y \in Y$ such that $f(x) = y$. However, notice that the converse *does not have to hold!* Let’s see why.

First, it is possible to have a function where two distinct inputs produce the same output. Formally, there might exist sets X and Y and a function $g : X \rightarrow Y$ such that:

$$(\exists x_1, x_2 \in X)((x_1 \neq x_2) \wedge (g(x_1) = g(x_2)))$$

For example, consider the function $g : \{0, 1, 2, 3\} \rightarrow \{0, 1\}$ given by the mapping below.

$$g(0) := 0 \quad g(1) := 1$$

$$g(2) := 0 \quad g(3) := 1$$

³The expression $f : X \rightarrow Y$ is a *complete sentence* that is usually read in English as “ f is a function from X to Y .”

Second, it is possible for there to be elements of the codomain of a function that *are not actually outputs* of that function. Formally, there might exist sets X and Y and a function $h : X \rightarrow Y$ such that the following sentence is satisfied.

$$(\exists y \in Y)(\forall x \in X)(h(x) \neq y)$$

For example, consider the function $h : \mathbb{N} \rightarrow \mathbb{N}$ given by $h(x) := x + 1$ for each $x \in \mathbb{N}$. Notice that $0 \in \text{cod}(h)$, but $(\forall x \in \text{dom}(h))(h(x) \neq 0)$. We will revisit these ideas soon.

7.2 Sequences and Iteration

Definition 7.3: Sequence.

sequence

Suppose that A is a set. We say that f is an *infinite sequence over A* $:\Leftrightarrow f : \mathbb{N} \rightarrow A$.

We say that f is a *finite sequence over A* $:\Leftrightarrow (\exists n \in \mathbb{N})(f : n \rightarrow A)$. Usually, if we call something a “sequence” without qualification, it is assumed we are referring to an *infinite* sequence; though, of course, it is best to be specific when possible.⁴

For example, the binary sequence “0, 1, 0, 1, 1, 0, 1, 1, 0, ...” is formally just a function $\beta : \mathbb{N} \rightarrow \{0, 1\}$ where $\beta(0) := 0$, and $\beta(1) := 1$, and $\beta(2) := 0$, and $\beta(3) := 1$, *et cetera*...

Definition 7.4: Iterated Addition.

Σ

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a sequence of natural numbers. We recursively define the *iterated sum* of finitely many values of f below for every $a \in \mathbb{N}$ and $b \in \mathbb{N}$.

$$\begin{aligned} \sum_{i=a}^b f(i) &:= 0 && \text{if } b < a \\ \sum_{i=a}^a f(i) &:= f(a) && \text{if } b = a \\ \sum_{i=a}^{\text{succ}(b)} f(i) &:= \left(\sum_{i=a}^b f(i) \right) + f(\text{succ}(b)) && \text{if } b > a \end{aligned}$$

Theorem 7.1.

For all $n \in \mathbb{N}$, we have that $2 \cdot \sum_{i=0}^n i = n \cdot (n + 1)$.

Proof. We will prove $(\forall n \in \mathbb{N})(2 \cdot \sum_{i=0}^n i = n \cdot (n + 1))$ by *weak induction*.

Basis Step:

Observe the following.

$$\begin{aligned} 2 \cdot \sum_{i=0}^0 i &= 2 \cdot 0 && \text{by definition of iterated addition} \\ &= 0 && \text{by definition of multiplication} \\ &= 0 \cdot 0 && \text{by definition of multiplication} \\ &= (0 \cdot 0) + 0 && \text{by definition of addition} \\ &= 0 \cdot \text{succ}(0) && \text{by definition of multiplication} \\ &= 0 \cdot 1 && \text{by definition of 1} \\ &= 0 \cdot (0 + 1) && \text{because } (\forall x \in \mathbb{N})(0 + x = x) \end{aligned}$$

⁴Note that we have *not yet defined* what “infinity” or “finiteness” are. We *do not* yet have a formal understanding of what it means for a set to be “infinite” or “finite” yet.

In the *basis step*, we need to prove $2 \cdot \sum_{i=0}^0 i = 0 \cdot (0 + 1)$.

Thus, we now know $2 \cdot \sum_{i=0}^0 i = 0 \cdot (0 + 1)$ as desired.

Inductive Step:

Let $k \in \mathbb{N}$, and assume $2 \cdot \sum_{i=0}^k i = k \cdot (k + 1)$. We can now observe the following.

$$\begin{aligned}
 2 \cdot \sum_{i=0}^{\text{suc}(k)} i &= 2 \cdot \left(\left(\sum_{i=0}^k i \right) + \text{suc}(k) \right) && \text{by definition of iterated addition} \\
 &= \left(2 \cdot \sum_{i=0}^k i \right) + (2 \cdot \text{suc}(k)) && \text{by distributivity of } \cdot \text{ over } + \\
 &= (k \cdot (k + 1)) + (2 \cdot \text{suc}(k)) && \text{by the inductive hypothesis} \\
 &= (k \cdot (k + 1)) + (2 \cdot (k + 1)) && \text{because } (\forall x \in \mathbb{N})(\text{suc}(x) = x + 1) \\
 &= ((k + 1) \cdot k) + ((k + 1) \cdot 2) && \text{by commutativity of multiplication} \\
 &= (k + 1) \cdot (k + 2) && \text{by commutativity of multiplication} \\
 &= (k + 1) \cdot (k + (1 + 1)) && \text{because } 1 + 1 = 2 \\
 &= (k + 1) \cdot ((k + 1) + 1) && \text{by associativity of addition} \\
 &= \text{suc}(k) \cdot (\text{suc}(k) + 1) && \text{because } (\forall x \in \mathbb{N})(\text{suc}(x) = x + 1)
 \end{aligned}$$

Thus, we have shown $2 \cdot \sum_{i=0}^{\text{suc}(k)} i = \text{suc}(k) \cdot (\text{suc}(k) + 1)$ as required.

Therefore, we can conclude $(\forall n \in \mathbb{N})(2 \cdot \sum_{i=0}^n i = n \cdot (n + 1))$.

QED

Definition 7.5: Iterated Multiplication.

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a sequence of natural numbers. We define the *iterated product* of finitely many values of f recursively below for every $a \in \mathbb{N}$ and $b \in \mathbb{N}$.

$$\begin{aligned}
 \prod_{i=a}^b f(i) &:= 1 && \text{if } b < a \\
 \prod_{i=a}^a f(i) &:= f(a) && \text{if } b = a \\
 \prod_{i=a}^{\text{suc}(b)} f(i) &:= \left(\prod_{i=a}^b f(i) \right) \cdot f(\text{suc}(b)) && \text{if } b > a
 \end{aligned}$$

□